PROG 7313 [POE PART 1]

PLANNING & DESIGN

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INTRODUCTION

The process of creating a personal budgeting application calls for a structured and user-oriented approach to ensure functionality, as well as usability. Based on previous studies of current budgeting apps, like Monefy, Spendee, and EasyBudget, it was found that most people stop using financial tracking apps because they are complex, lack interest, and have a poor intuitiveness. These findings provide a direct input into planning and design of the proposed application, while ensuring that the application addresses common usability issues and meets the functional requirements as defined in the project brief. (Dribbble.com, 2026)

This document shows the planning and design phase of the CashaNova budgeting application. It covers the overview of the application, detailed functional requirements, user interface design, and project planning. The objective is to design an application that runs on mobile devices and is simple, engaging, and efficient, aiding users in tracking their expenditures, managing budgets and in attaining financial ambitions. This design matches the needs defined in the assessment brief (key features in terms of expenses, categories, budget goals, gamification, etc.). In addition, the document is used to ensure that the application design is practical enough to be implemented yet innovative and user-centred at the same time. (Alenazi and Sas, 2023)

Brief Application Overview

Name Of Application: CashaNova

Application Icon:

INNOVATIVE FEATURES

The proposed application is called CashaNova and is a modern and interesting budgeting application, made for making financial management easier. The name represents a new and innovative way of doing personal finance, and it targets a younger, tech-savvy audience. (Romy Fleur Willemsen et al., 2023)

CashaNova's goal is to offer a simple yet powerful platform for users to track their daily expenses, manage their monthly budget, monitor their financial habits, and reach their financial savings goals. The application limits the stress associated with financial management by making the experience interactive, visually engaging and easy to understand for users with all experience levels. (Romy Fleur Willemsen et al., 2023)

Key features are expense tracking, category management, budget goals, receipt uploads, recurring transactions and visual dashboards, elements of gamification, Chat Buddy assistant, and data export functionality. These features make it, thus, functional and appealing as well, hence ensuring long-term retention and consistency of use over time. (Romy Fleur Willemsen et al., 2023)

CashaNova Budget Management Application – Detailed Feature Requirements

The application needs to support user registration and authentication. Users have to create secure accounts by using personal information like name, email, username, and password. Input validation is used to ensure the accuracy and security of data. (văn bằng trần, 2024)

Passwords are encrypted before they are even stored and login functionality is used to verify passwords before allowing access to the dashboard. (Zhang et al., 2022)

The dashboard is the central place of the application. It shows current balance, income, expenses as well as budget progress. It also highlights overspending categories in a visual way and provides navigation to all other features that ensures that the user can quickly access the information they need. (Zhang et al., 2022)

Budget management enables the users by creating and maintaining monthly budgets by entering the income and fixed expenses. The system automatically computes what remains and assists users in adapting spending habits accordingly to better discipline finances. (Romy Fleur Willemsen et al., 2023)

Category management allows the user to create, edit and delete categories like groceries, transport, entertainment etc. This helps users to organise their finances well and get better insights about their spending patterns. (To, 2026)

Category spending limits are for users that want to set some maximum spending amount per category. The system monitors spending and sends alerts for limit breach or near-limit breach which helps users to stay within their financial goals. (To, 2026)

Expense tracking is a basic feature that enables users to record expenses with information such as description, amount, date, and category. The system automatically updates the balances and stores data securely to make the system reliable. (To, 2026)

Receipt upload functionality to enable users to upload proof of purchases in formats such as JPG, PNG and PDF. This increases record keeping and adds transparency for the users who want to review their spending. (To, 2026)

Recurring Expenses Users can use recurring expenses to automate regular transactions such as rent or subscriptions. This allows for less manual input and allows for increased efficiency while ensuring that regular expenses are never forgotten. (To, 2026)

Expense history and filtering features to allow users to view, search, edit, and delete past transactions based on date, category, or description allowing them to analyse financial behaviour more easily. (To, 2026)

Visual data representation involves graphs, summaries, etc., which help users make sense of spending trends and financial behaviour at a fast and effective pace. (To, 2026)

Gamification features like badges, streaks and challenges to motivate user to have habit of budgeting and enhance engagement. (To, 2026)

The Chat Buddy assistant offers financial advice, reminders, and suggestions to better engage with users and make them more aware of their finances, aiding as a supportive tool to the users. (To, 2026)

All data is securely stored with the use of local databases, encryption, and authentication measures to secure user data and keep it private. (To, 2026)

User Interface Design

1. Introduction

The application interface is designed to be simple, intuitive, aesthetically pleasing and easy to navigate. The login screen contains spaces for entering username and password, and places for registering and resetting passwords. (Milos Kujundzic, 2019)

2. Login Screen

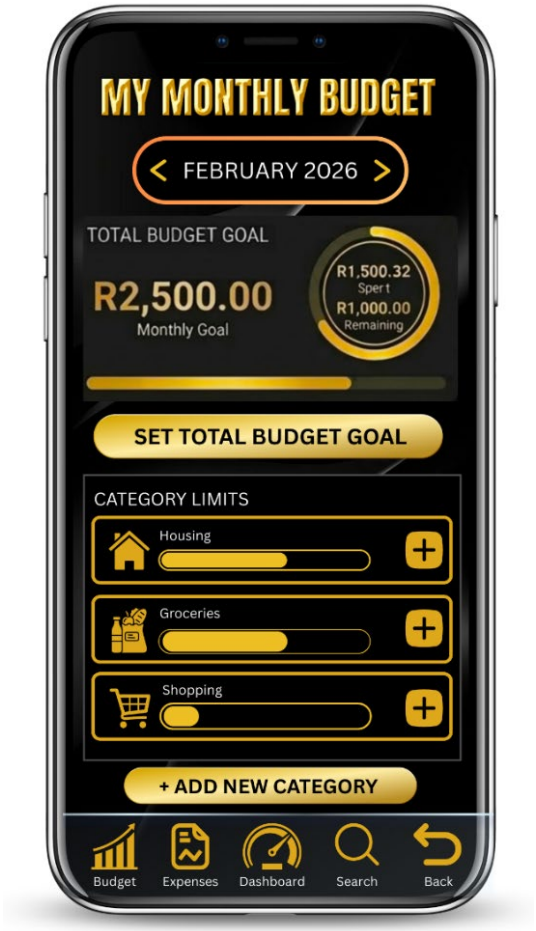
It will be where you will be safe to log in and view all personal finance information. It welcomes the user with a personalised name for the user. This screen has the following fields: username field, password field, and the login button, links to forgotten password, forgotten username and a register now link that one may use in case he or she is a new user. Upon filling the right creds and clicking on the login, you are automatically redirected to the dashboard. The Register Now button will take one to registration page and links with password/username initiate account recovery process. This is in effect where one enters the app. (Milos Kujundzic, 2019)

3. Registration Screen

The registration screen is where new users can set up accounts with the system by typing in their personal information. On successful registration, the users are taken to the login screen. (To, 2026)

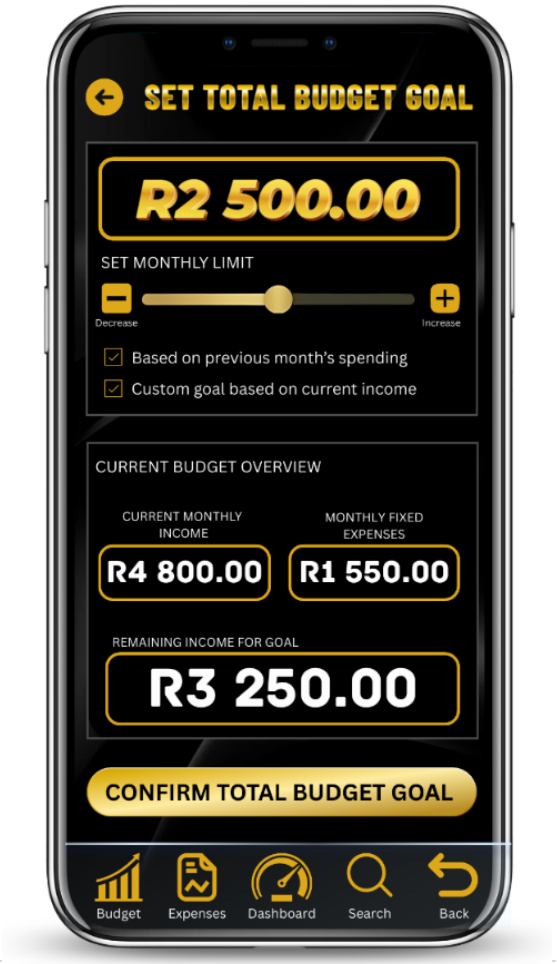
4. Dashboard Screen

The dashboard screen gives an overview of financial information and serves as the primary navigation point, with users able to quickly reach the main features on the screen. Expenses Users can use recurring expenses to automate regular transactions such as rent or subscriptions. The Chat Buddy assistant offers advice based on fixed prompts relating to budgeting. (Milos Kujundzic, 2019)



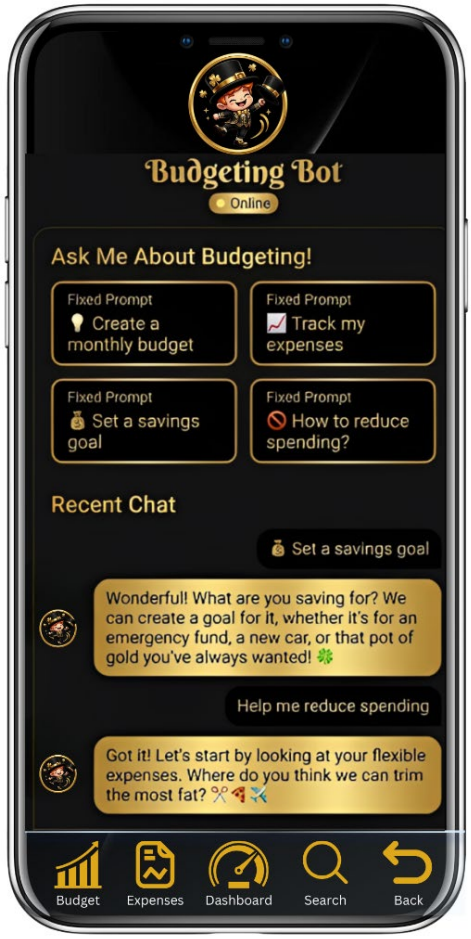
5. Budget Screen

The budget screen enables users to manage the income, expenses, and category limits. Users can add and edit categories in this screen. (To, 2026)



6. Set Total Budget Goal

Allows user to set monthly budget target and provides a budget overview of current monthly income vs. expenses and remaining amount to budget (To, 2026)



7. Budgeting Bot

Allows user to chat and interact with an AI Bot that allows user to query fixed prompts and guidance/ advice will be provided by the bot. (Milos Kujundzic, 2019)

8. Inbox

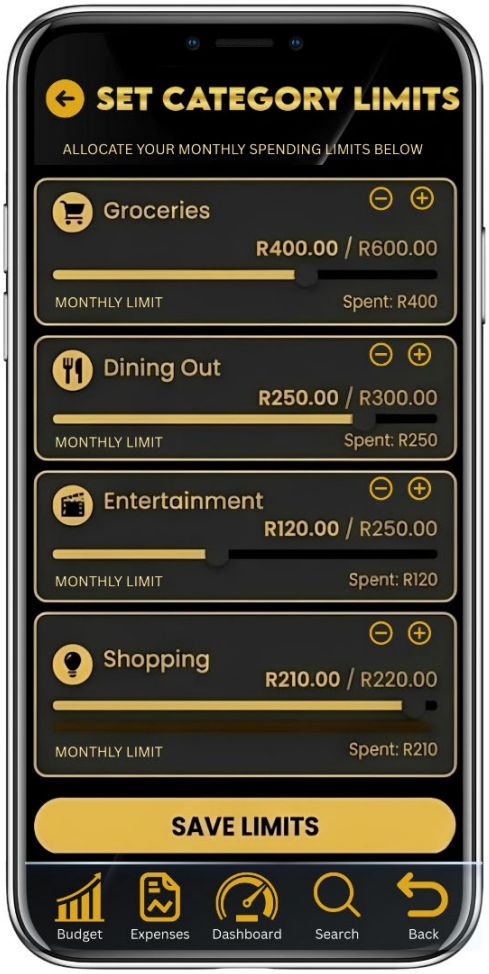
Gamification features like badges, streaks and challenges to motivate user to have habit of budgeting and enhance engagement. (Milos Kujundzic, 2019)

9. Category Creation Screen

The category creation screen allows users to customise categories with names, icons and colours, which makes the interface more personalised. (Milos Kujundzic, 2019)

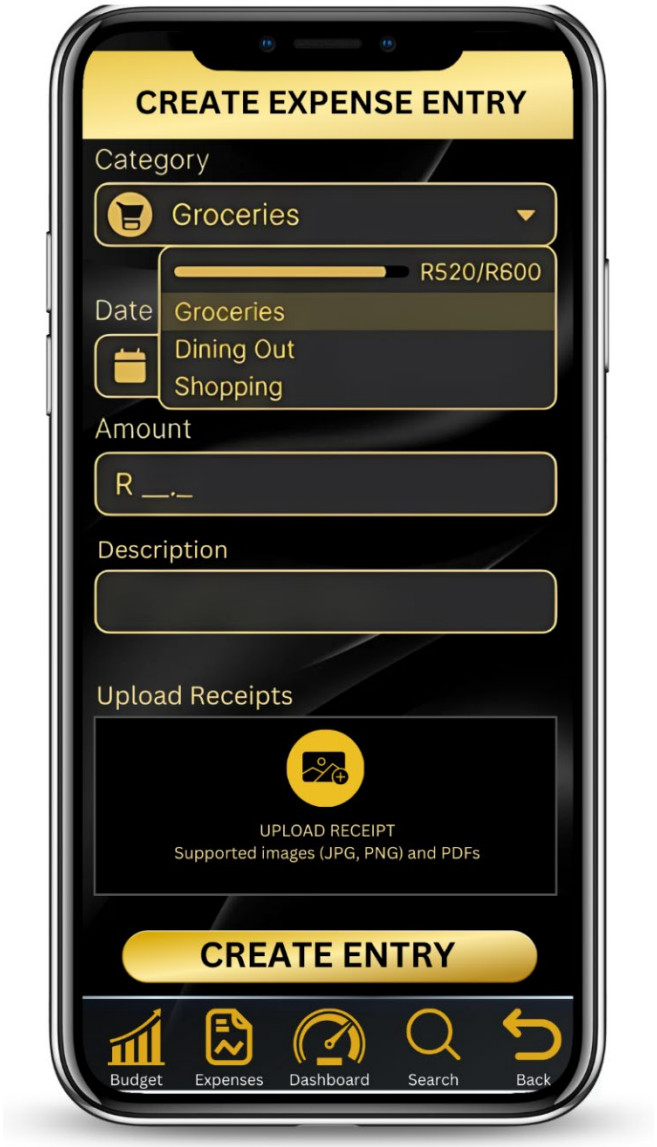
10. Set Category Limit

Allows users to limit spending on specific categories. (To, 2026)



11. Expense Screen

The expense screen will show a list of transactions and enable respondents to enter new expenses or manage ongoing expenses. (To, 2026)



12. Create Expense Entry Screen

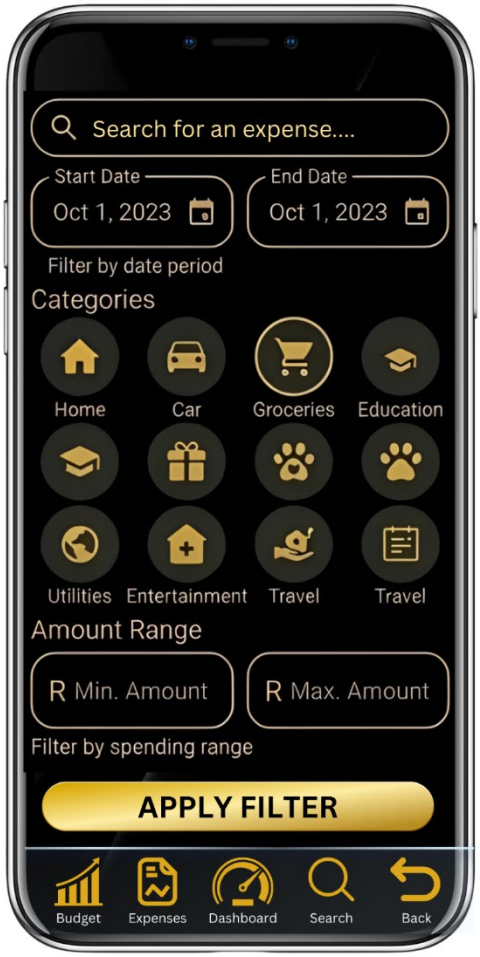
The create expense screen lets the users to publish the details of expenses and also upload receipts easily (To, 2026)

13 . Export Expenses Data

Exports expense data over a selected period or category as a csv. file. (Milos Kujundzic, 2019)

14. Setup Recurring Expenses

Allows user to setup expenses that occur on a consistent basis such as rent or insurance, this is so they do not have to keep recording the same expense frequently. (Milos Kujundzic, 2019)



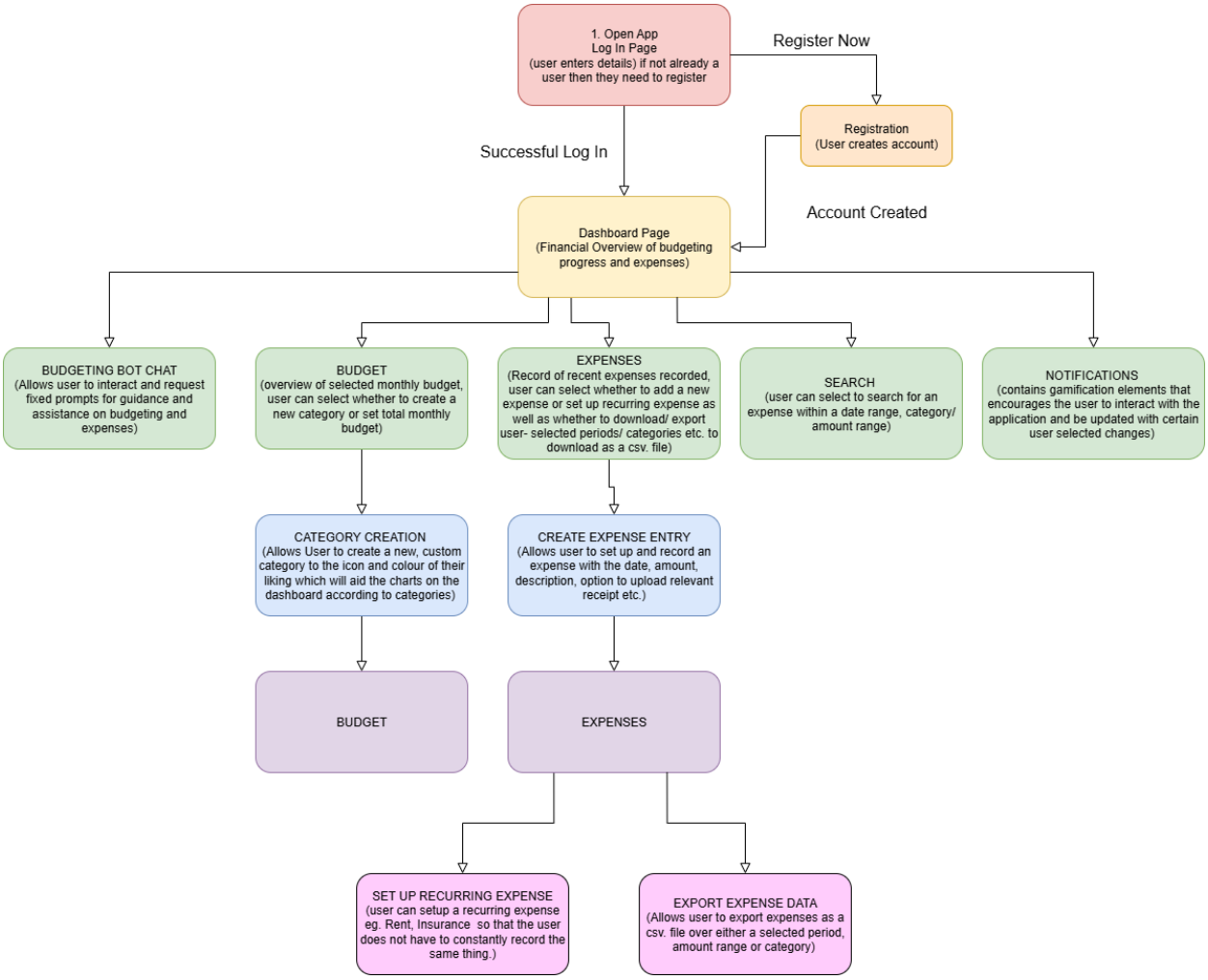
15. Search Screen

The search screen enables users to search for transactions based on different filters such as date, category, or description which provides better accessibility of data. (Milos Kujundzic, 2019)

10. Navigation Structure

The navigation structure is very simple and intuitive with the dashboard as the central hub. Users can easily switch between the screens with the help of a navigation menu so that the user has a smooth journey. (To, 2026)

Navigation Diagram:



Project Plan:

3 images included: IMAGE 1. Project Plan and Gantt Chart
IMAGE 2. Project Plan close-up
IMAGE 3. Gantt Chart close-up

images will also be included in submission for better view

IMAGE 1:

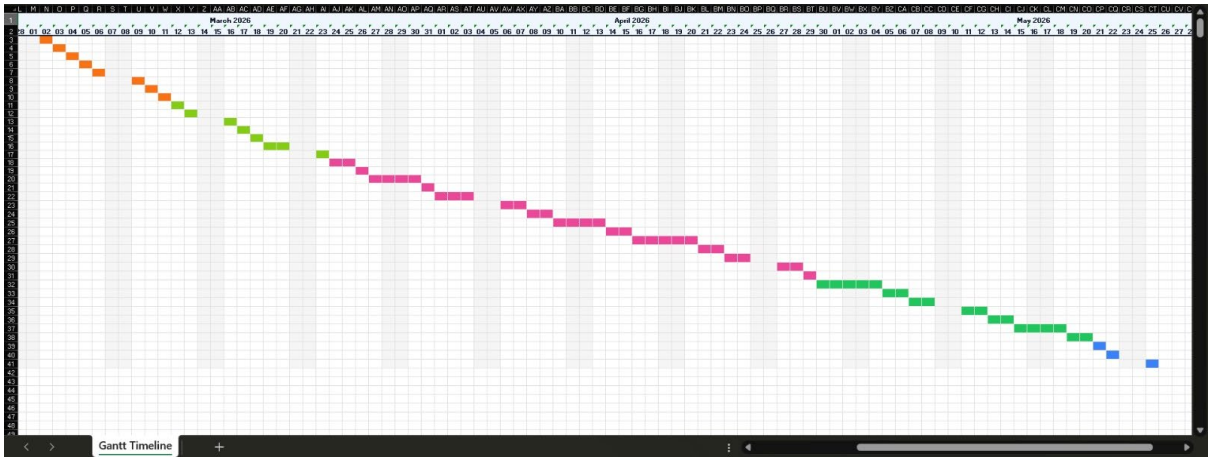
Workstream	Task Name	Assignees	Start Date	End Date	Duration
Part 1 Research	Identify 3 budgeting apps Research App 1 features Capture App 1 screenshots Research App 2 features Capture App 2 screenshots Research App 3 features	Research Team	2026-03-02	2026-03-02	1
Part 1 Research		Research Team Research Team Research Team Research Team Research Team	2026-03-03	2026-03-03	1
Part 1 Research			2026-03-04	2026-03-04	1
Part 1 Research			2026-03-05	2026-03-05	1
Part 1 Research			2026-03-06	2026-03-06	1
Part 1 Research			2026-03-09	2026-03-09	1
Part 1 Research	Create apps comparison infographi Write research findings document	Research Team Research Team	2026-03-10	2026-03-10	1
Part 1 Research			2026-03-11	2026-03-11	1
Planning and Design	Define app concept name Design application icon Define innovative features	Design Team	2026-03-12	2026-03-12	1
Planning and Design		Design Team	2026-03-13	2026-03-13	1
Planning and Design		Design Team	2026-03-16	2026-03-16	1
Planning and Design	Write functional requirements Define database entities Design UI mockups Create project planning docs Setup Android Studio project Setup GitHub repository	Design Team	2026-03-17	2026-03-17	1
Planning and Design	Design application UI layout Setup RoomDB database Implement user registration	Design Team	2026-03-18	2026-03-18	1
Planning and Design		Design Team	2026-03-19	2026-03-20	2
Planning and Design		Design Team	2026-03-23	2026-03-23	1
Prototype Development		Development Team	2026-03-24	2026-03-25	2
Prototype Development		Development Team	2026-03-26	2026-03-26	1
Prototype Development		Development Team	2026-03-27	2026-03-30	4
Prototype Development		Development Team	2026-03-31	2026-03-31	1
Prototype Development		Development Team	2026-04-01	2026-04-03	3
Prototype Development	Implement login authentication	Development Team	2026-04-06	2026-04-07	2
Prototype Development	Implement expense categories	Development Team	2026-04-08	2026-04-09	2
Prototype Development	Implement add expense screen Create expense list display	Development Team	2026-04-10	2026-04-13	4
Prototype Development		Development Team	2026-04-14	2026-04-15	2
Prototype Development	Implement budget calculations Add spending goals feature Conduct unit testing Debug and fix bugs Improve UI optimization	Development Team	2026-04-16	2026-04-20	5
Prototype Development	Add charts analytics Implement dashboard screen Add gamification features Improve UI styling	Development Team QA Team	2026-04-21	2026-04-22	2
Prototype Development	Add extra features Conduct full testing Fix final bugs Prepare APK build Record demo video		2026-04-23	2026-04-24	2
Prototype Development		Development Team	2026-04-27	2026-04-28	2
Prototype Development		Development Team	2026-04-29	2026-04-29	1
Final Development		Development Team	2026-04-30	2026-05-04	5
Final Development		Development Team	2026-05-05	2026-05-06	2
Final Development		Development Team	2026-05-07	2026-05-08	2
Final Development		Development Team	2026-05-11	2026-05-12	2
Final Development		Development Team QA Team	2026-05-13	2026-05-14	2
Final Development			2026-05-15	2026-05-18	4
Final Development		Development Team	2026-05-19	2026-05-20	2
Final Preparation		Development Team Media Team	2026-05-21	2026-05-21	1
Final Preparation			2026-05-22	2026-05-22	1
Final Preparation	Complete final documentation	Documentation Team 2026-05-25		2026-05-25	1

IMAGE 2:

	A	B	C	D	E	F
2	Workstream	Task Name	Assignees	Start Date	End Date	Duration
3	Part 1 Research	Identify 3 budgeting apps	Research Team	2026-03-02	2026-03-02	1
4	Part 1 Research	Research App 1 features	Research Team	2026-03-03	2026-03-03	1
5	Part 1 Research	Capture App 1 screenshots	Research Team	2026-03-04	2026-03-04	1
6	Part 1 Research	Research App 2 features	Research Team	2026-03-05	2026-03-05	1
7	Part 1 Research	Capture App 2 screenshots	Research Team	2026-03-06	2026-03-06	1
8	Part 1 Research	Research App 3 features	Research Team	2026-03-09	2026-03-09	1
9	Part 1 Research	Create apps comparison infograp	Research Team	2026-03-10	2026-03-10	1
10	Part 1 Research	Write research findings documer	Research Team	2026-03-11	2026-03-11	1
11	Planning and Design	Define app concept name	Design Team	2026-03-12	2026-03-12	1
12	Planning and Design	Design application icon	Design Team	2026-03-13	2026-03-13	1
13	Planning and Design	Define innovative features	Design Team	2026-03-16	2026-03-16	1
14	Planning and Design	Write functional requirements	Design Team	2026-03-17	2026-03-17	1
15	Planning and Design	Define database entities	Design Team	2026-03-18	2026-03-18	1
16	Planning and Design	Design UI mockups	Design Team	2026-03-19	2026-03-20	2
17	Planning and Design	Create project planning docs	Design Team	2026-03-23	2026-03-23	1
18	Prototype Development	Setup Android Studio project	Development Team	2026-03-24	2026-03-25	2
19	Prototype Development	Setup GitHub repository	Development Team	2026-03-26	2026-03-26	1
20	Prototype Development	Design application UI layout	Development Team	2026-03-27	2026-03-30	4
21	Prototype Development	Setup RoomDB database	Development Team	2026-03-31	2026-03-31	1
22	Prototype Development	Implement user registration	Development Team	2026-04-01	2026-04-03	3
23	Prototype Development	Implement login authentication	Development Team	2026-04-06	2026-04-07	2
24	Prototype Development	Implement expense categories	Development Team	2026-04-08	2026-04-09	2
25	Prototype Development	Implement add expense screen	Development Team	2026-04-10	2026-04-13	4
26	Prototype Development	Create expense list display	Development Team	2026-04-14	2026-04-15	2
27	Prototype Development	Implement budget calculations	Development Team	2026-04-16	2026-04-20	5
28	Prototype Development	Add spending goals feature	Development Team	2026-04-21	2026-04-22	2
29	Prototype Development	Conduct unit testing	QA Team	2026-04-23	2026-04-24	2
30	Prototype Development	Debug and fix bugs	Development Team	2026-04-27	2026-04-28	2
31	Prototype Development	Improve UI optimisation	Development Team	2026-04-29	2026-04-29	1
32	Final Development	Add charts analytics	Development Team	2026-04-30	2026-05-04	5
33	Final Development	Implement dashboard screen	Development Team	2026-05-05	2026-05-06	2
34	Final Development	Add gamification features	Development Team	2026-05-07	2026-05-08	2
35	Final Development	Improve UI styling	Development Team	2026-05-11	2026-05-12	2
36	Final Development	Add extra features	Development Team	2026-05-13	2026-05-14	2
37	Final Development	Conduct full testing	QA Team	2026-05-15	2026-05-18	4
38	Final Development	Fix final bugs	Development Team	2026-05-19	2026-05-20	2
39	Final Preparation	Prepare APK build	Development Team	2026-05-21	2026-05-21	1
40	Final Preparation	Record demo video	Media Team	2026-05-22	2026-05-22	1
41	Final Preparation	Complete final documentation	Documentation Team	2026-05-25	2026-05-25	1

Gantt Timeline

IMAGE 3:



Conclusion

The CashaNova budgeting application planning and design have been properly developed based on the research and project needs. The design is characterised using usability, simplicity and focused aspects of user experience such that the user is able to effectively manage their finances, without being overburdened. (Savitha and Hawaldar, 2022)

By adding essential features like expense tracking, creating a budget and visualizing data, combined with more innovative features like gamification and the Chat Buddy assistant, the application offers functionality and user engagement. (Savitha and Hawaldar, 2022)

Overall, this design forms a solid foundation for development and is designed to ensure that the end application not only meets academic requirements but also real-world user requirements, as well as being practical, scalable, and easy to use. (Savitha and Hawaldar, 2022)

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